

Structure of White Paper:

Abstract (300 words or less)

Introduction: describe your results and how you add to the existing literature (3 pages or so 1.5 spacing)

Lit review: 1 to 2 pages max

Empirical Design (very long 15 to 20 pages)

All the rest below is subsections of the Empirical design

1. Describe the model design and setup

2. Describe the training of the AI models – what information where they fed before they were set to transact

What utility functions were used (ideally we get the same results whether we use CRRA utility or CARA or Epstein zin etc)

Each Agent is given a utility function = $U(W)$ where W is final wealth and they are told to sell a good with another agent and can select the coins they want to use

And we tell the agent to try and maximize $U(W)$ over a finite amount of time (transactions with other agents)

1. Baseline model

Terms below

- Governance features (what is the asset backed by) – “High governance” denotes Genius act compliant
- What does it track (medium of exchange) - USD or Euro, gold etc
- Liquidity on an exchange (bid-ask spread) – “High Liquidity” denotes low bid-ask spread
- Volatility of the stablecoin - how close does it stay to a dollar
- What exchange it trades on: gas fees associated with it

2. Verification of baseline results

We need to first talk about how AI agents are assumed to like the following

- High governance coins over low governance coins
- High liquidity coins over low liquidity coins

- Low volatility over high volatility
- Low gas fees over high gas fees

And our initial tests verify these results.

Next while not entirely evident we first test preferences over medium of exchange. We verify that all agents no matter what their risk aversion prefer USD medium coins over Euro.

Following this we go into our hypotheses:

H1: The more risk averse the agent, the more they prefer USD over Euro over gold

- (Aside: we should note that all simulations prefer USD, no matter the level of risk aversion, but the risk averse agent prefers it the most. And only the risk seeking agent wants gold)

Then we head into the rest of our hypotheses:

H2: The more risk averse the agent, the more the agent prefers High governance coins over concerns of the medium of exchange (USD v Euro v gold)

H3: The more risk averse the agent, the more the agent prefers High governance coins over Highly Liquid coins

H4: The more risk averse the agent, the more the agent prefers High governance coins over low volatility coins

H5: The more risk averse the agent, the more the agent prefers High governance coins over low gas fees (the cost of the rails it trades on)

H6: The more risk averse the agent, the more the agent cares about the medium of exchange (USD v Euro) over the liquidity of the coin

H7: The more risk averse the agent, the more the agent cares about the medium of exchange (USD v Euro) over the volatility of the coin

H8: The more risk averse the agent, the more the agent cares about the medium of exchange (USD v Euro) over the gas fees of the coin

H9: The more risk averse the agent, the more the agent cares about liquidity of the coin over the volatility of the coin

H10: The more risk averse the agent, the more the agent cares about liquidity of the coin over the gas fees associated with the coin

H11: The more risk averse the agent, the more the agent cares about the volatility of the coin over the gas fees associated with the coin

Tests of initial hypotheses (important – we need tables like below to show results)

Example of a table

H1:

Risk Aversion Level	CRAA – Equilibrium holdings	CARA – Equilibrium Holdings	Epstein Zin
Risk Neutral ($a=0$)	50% USD, 30% Euro, 20% gold	50% USD, 30% Euro, 20% gold	50% USD, 30% Euro, 20% gold
Moderate risk averse ($a=2$)			
More Risk averse ($a=4$)			
Most Risk Averse ($a=6$)	80% USD, 20% Euro	75% USD, 25% Euro	85% USD, 15% Euro

Next to test our next hypotheses, we develop an equivalence framework.

Where we give the AI agents the option of receiving a High governance coin with some bid-ask spread (X) or a low governance coin with some other bid-ask spread (Y). In our simulations we set $Y = 0.01\%$ and see where the AI agent becomes indifferent by Varying X.

For instance, we see that in the CRAA preference case, when $a=0$, we find that $X=0.02\%$. This means $X-Y = 0.01\%$

In other words, the AI agent will accept a 0.01% higher bid-ask spread if given the option of switching to a high governance coin

Example of Result Tables:

H3:

Risk Aversion Level	CRAA – Switch from High Gov coins to Low gov coins (equivalent change in bid-ask needed for indifference) [X-Y]	CARA – Switch from High Gov coins to Low gov coins (equivalent change in bid-ask needed for indifference)) [X-Y]	Epstein Zin– Switch from High Gov coins to Low gov coins (equivalent change in bid-ask needed for indifference)) [X-Y]
Risk Neutral ($a=0$)	0.01% (example)		
Moderate risk averse ($a=2$)			

More Risk averse (a=4)			
Most Risk Averse (a=6)	0.04% (example)		

If we wanted to the AI agent to switch from a High Gov coin to a low Gov coin, how much would we need to compensate them.

H4:

Risk Aversion Level	CRAA – Switch from High Gov coins to Low gov coins (equivalent change in volatility needed for indifference) [X-Y]	CARA – Switch from High Gov coins to Low gov coins (equivalent change in volatility needed for indifference)) [X-Y]	Epstein Zin– Switch from High Gov coins to Low gov coins (equivalent change in volatility needed for indifference)) [X-Y]
Risk Neutral (a=0)	.02% (example)		
Moderate risk averse (a=2)			
More Risk averse (a=4)			
Most Risk Averse (a=6)	.06% (example)		

If I give the AI agent a High Governance coin with a volatility of .10%. What reduction in volatility would induce him to switch to the low governance coin and be indifferent between the two. In the above risk neutral case it would take lowering the volatility of the coin down to 0.08% to get them to switch to a low vol coin (and therefore the answer is 0.02% = 0.10-0.08).

For the risk averse agent they would only switch if we reduced the volatility of the coin from 0.10 to 0.04 = 0.06%

2. Cross Border transactions Change

Cross border – how do we model agents on different sides of the Atlantic (one that cares about Euros and one that cares about USD)

- One agent maxes wealth in euros, one maxes wealth in USD, each must pay a transaction cost to move their money from USDC to Euro (or from EURC to USD) – small % tax to convert money if the money is not the type that they want

Example Table:

Risk Aversion Level	CRAA – Equilibrium holdings	CARA – Equilibrium Holdings	Epstein Zin
Risk Neutral ($a=0$)	50% USD, 30% Euro, 20% gold	50% USD, 30% Euro, 20% gold	50% USD, 30% Euro, 20% gold
Moderate risk averse ($a=2$)			
More Risk averse ($a=4$)			
Most Risk Averse ($a=6$)	80% USD, 20% Euro	75% USD, 25% Euro	85% USD, 15% Euro

Focus on Maybe just H2, H6, H7, H8

3. Event based analysis

Don't have to do all hypotheses again here but we run through a depeg event and see how the attitudes of the agents change.

Run through a financial crisis and do some hypotheses here too.

(Maybe 4 tables in this section)

The research project focuses on five main variables: the governance features of a coin, the medium of exchange, the liquidity of the coin, the volatility of the coin, and the gas fees associated with it. In the first section of the research paper, the team should explain how the model verifies the assumption that AI agents prefer high-governance coins over low-governance coins, highly liquid coins over low-liquidity coins, low-volatility coins over high-volatility coins, and coins with low gas fees over those with high gas fees. High-governance coins are described as coins that are compliant with the Genius Act. These initial tests are

primarily intended to verify that the model behaves according to these expected preferences.

The first major hypothesis examines risk aversion. Specifically, the hypothesis states that the more risk-averse an agent is, the more the agent will prefer USD over Euro and gold. The researchers can test this by running agents with different levels of risk aversion and observing their preferred holdings. The agents could be given a utility function and allowed to continue trading until they reach their preferred equilibrium. For example, a risk-neutral group of agents might eventually hold a hypothetical portfolio consisting of 50% USD, 30% Euro, and 20% gold. More risk-averse agents would be expected to reach a different preferred equilibrium. The agents would be allowed to freely select which currencies or coins they want to use.

The model will use a one-year, or 365-day, timescale. The empirical design should clearly describe the model setup and training process, including exactly what information was provided to the AI agents and what utility functions they were given. Each agent is assigned a utility function, represented generally as $U(W)$. The paper should explain the model setup in detail so that readers understand how the agents were constructed, what information they received, and how they made decisions.

The paper is expected to contain a significant amount of writing as well as tables and potentially graphs. The baseline model may contain approximately 10 or 11 hypotheses, with a corresponding table for each hypothesis. The researchers will then move into a cross-border analysis, where they will select the most important hypotheses, followed by an event-based analysis. The professor estimated that there could be approximately 20 small tables throughout the paper, although graphical representations could be used instead when appropriate. The literature review is expected to be approximately one to two pages, while the empirical design section will be longer because it must describe the model, the information provided to the AI agents, the utility functions, and the data sources. The paper should also contain a section explaining the different features that a stablecoin can have.

For the cross-border analysis, Hypothesis 1 was identified as especially important. Hypotheses 2, 6, 7, and 8 were also selected because they focus on the interaction between coin type and different levels of risk aversion. These hypotheses are considered particularly relevant to understanding how different types of coins interact with risk preferences in cross-border transactions.

A major part of the proposed research design is the development of an equivalence framework. Under this framework, AI agents would receive a high-governance coin with one

bid-ask spread and a low-governance coin with another bid-ask spread. The researchers would fix one bid-ask spread and vary the other until the agent becomes indifferent between the two coins. For example, if the high-governance coin has a bid-ask spread of 0.01 and the agent becomes indifferent when the other coin reaches 0.02, the difference between the two spreads represents the compensation necessary for the agent to switch. This approach is intended to measure how much an AI agent values a particular characteristic, such as high governance, by determining what economic benefit would be required for the agent to give up that characteristic.

This equivalence framework can be applied to each hypothesis. For example, when testing a hypothesis involving governance, the researchers could determine how much compensation would be required for an AI agent to switch from a high-governance coin to a low-governance coin. The compensation could be represented through a better bid-ask spread. The professor suggested that a risk-neutral agent might require only a 0.01% improvement in the bid-ask spread, while a more risk-averse agent might require a 0.04% improvement. These numbers were presented only as hypothetical examples, and the actual results would need to come from the simulations.

The same framework can be applied to volatility. In this case, the researchers would examine how much volatility would have to decrease before an agent would switch from a high-governance coin to a low-governance coin. For example, if an agent initially holds a high-governance coin with a volatility of 0.10%, the researchers could gradually reduce the volatility and determine the point at which the agent becomes indifferent between the high-governance and low-governance alternatives. In the example discussed, reducing volatility from 0.10% to 0.08% would represent a 0.02% reduction required for the risk-neutral agent to switch.

The goal is to isolate one factor at a time for each hypothesis. For example, the researchers could compare a high-governance coin with a particular level of volatility against a low-governance coin with a different level of volatility. They would then determine the point at which agents would be willing to switch between the two. This approach is intended to provide a cleaner and more academically rigorous way of measuring the importance of individual coin characteristics.

The proposed experimental setup would involve placing AI agents in a sandbox where they can conduct transactions using coins with different characteristics. For a particular hypothesis, the researchers would isolate two variables, such as governance and volatility, and allow the agents to transact and learn about the coins. After the agents have completed their transactions and learned about the different coins, the researchers would ask them whether they would switch from one type of coin to another under a specified

change in the relevant characteristic. The professor suggested that this question could be asked at the end of the 365-day simulation rather than after every transaction.

The expected results would show that more risk-averse agents require a greater change in a coin's characteristics before they are willing to switch. For example, a more risk-averse agent might only switch from a high-governance coin to a low-governance coin if the volatility were reduced by 0.06%, such as from 0.10% to 0.04%. The same general approach would then be applied to the other hypotheses. Hypothesis 5 would focus on changes in gas fees, while another hypothesis would examine USD, Euro, and gold with different levels of liquidity and determine how much liquidity would need to change before an agent switches from one medium of exchange to another.

The researchers discussed creating separate sandboxes for each hypothesis so that the relevant variables could be isolated. For example, one sandbox could contain 100 agents transacting with coins that differ in governance and liquidity characteristics, while another could contain coins that differ in governance and volatility. Another sandbox could examine USD, Euro, and gold with different liquidity levels. Overall, the proposed design would involve approximately 11 different sandboxes corresponding to the different hypotheses.

The main computational limitation is not necessarily the amount of computation itself but the time required for the API-based LLM calls. Running the models through an API takes significant time, and running approximately 100 models locally would require substantial server resources. The team had already encountered a coding roadblock, but it had been fixed. They expected that implementing the new approach would require modifications to the existing code, but believed it was feasible because the three researchers could divide the simulations among themselves and run multiple simulations simultaneously.

The final paper will contain tables corresponding to the different hypotheses, along with additional tables for the cross-border and event-based sections. The professor emphasized that the equivalence results would be particularly interesting because they provide an economic interpretation of the agents' preferences. Rather than simply showing that an agent prefers one type of coin over another, the researchers can determine how much compensation, such as a better bid-ask spread or lower volatility, would be required to convince the agent to switch. This provides a more quantitative and academically rigorous way to present the results.